

Possible and Necessary Popularity in House Allocation PseudoCode

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1 Hypothesis

- Agents give their partial strict linear order. They indicate positions of each object in their ranking.
- There are as many objects as agents.
- We consider only strict linear order.

2 Properties

Proposition 1 (Verify Possibly popular matching). *If $\exists i, j \in N$ such that $\sigma(j)P_i\sigma(i)$ and $\exists o \in O$ such that $o \succ_j \sigma(j)$ then return False (the matching is not popular).*

Proposition 2 (Find Possibly popular completion). *If $\exists i, j \in N$ such that $\sigma(j)P_i\sigma(i)$ for $o \in O$ such that $(\sigma(j), o) \notin P$ add $(\sigma(j), o)$ in $\succ_j$*

Necessarily popular matching

Proposition 3 (Find Necessarily popular matching). *If $\exists i, j \in N$ such that $\sigma(j)P_i\sigma(i)$ and $o \in O$*

- $oP_j\sigma(j)$ or
- $(\sigma(j), o) \notin P_j$

Return False (it is not a necessarily popular matching).

Proposition 4 (Find Necessarily popular completion). *If $\exists i, j \in N$ such that $(\sigma(i), \sigma(j)), (\sigma(j), \sigma(i)) \notin P_i$ and $o \in O$ such that*

- $oP_j\sigma(j)$
- $(\sigma(j), o) \notin P_j$

Return False (it is not a necessarily popular matching).

Proposition 5. *If at least half of agents have the same fpost, it is not possible to find a popular matching.*

3 Algorithms

Algorithm 1 IsPreferredTo

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1: Input:  $o \in O, o' \in O, P_k \in P$ 
2: Output: boolean indicating either  $o$  is preferred to  $o'$  by agent  $k$ 
3:
4: for condition do
5:   operation
6: end for
7: return result

```

Algorithm 2 VerifPossiblePopularMatching

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1: Input:  $O$  the set of  $n$  objects,  $N$  the set of  $n$  agents,  $P = (P_1, \dots, P_n)$  the partial
   profile such that  $P_i = [(o_l, o_k), \dots]$  is the list of pairwise partial preference of agent  $i$ ,
    $\mu = \{1: \sigma(1), \dots, n: \sigma(n)\}$  is the matching ( $\sigma(i)$  is the assigned object to  $i$  in the matching).
2: Output: boolean indicating either the matching  $\mu$  is possibly popular
3: for each agent  $i \in N$  do
4:   for each preference  $(o_{sup}, o_{inf}) \in P_i$  do
5:     NotFpost  $\leftarrow o_{inf}$  is  $\sigma(i)$ 
6:     NotDominatedByFpost  $\leftarrow False$ 
7:     for each preference  $(o, o') \in P_j$  do
8:       NotDominatedByFpost  $\leftarrow o_{sup}$  is  $o'$ 
9:     end for
10:    if NotFpost and NotDominatedByFpost then
11:      return False
12:    end if
13:  end for
14: end for
15: return True

```

Algorithm 3 VerifPossiblePopularMatching

```
1: Input:  $O$  the set of  $n$  objects,  $N$  the set of  $n$  agents,  $P = (P_1, \dots, P_n)$  the partial profile
   such that  $P_i = [?, \sigma(i), o, \dots]$  is the list of partial preference of agent  $i$  (? indicates unknown
   preference and  $\sigma(i)$  is the assigned object to  $i$  in the matching),  $\mu = \{1: \sigma(1), \dots, n: \sigma(n)\}$  is
   the matching.
2: Output: boolean indicating either the matching  $\mu$  is possibly popular,  $P$  a completion of the
   partial profile making  $\mu$  possibly popular.
3:
4: for each agent  $i \in N$  do
5:   for each preference  $p \in P_i$  do
6:     if  $p$  is unknown (equal to ?) then
7:       // Is there a completion such as  $p$  is the fpost matched to another agent in  $\mu$ ?
8:       if  $\sigma(i)$  not in  $P_i$  then
9:         Assign  $\sigma(i)$  to  $p$ 
10:      else
11:        Find a matched fpost (should not be a fpost for more than half agents) and
        assigned it to  $p$ 
12:      end if
13:    else if  $p$  is  $\sigma(i)$  then
14:      Complete the preference list  $P_i$  with the remaining objects.
15:      Break
16:    else
17:      //  $p$  is an object with a better rank than  $\sigma(i)$ 
18:      if  $p$  is a fpost matched to another agent then
19:        Continue
20:      else
21:        //  $p$  is not a matched fpost, the matching  $\mu$  cannot be popular
22:        return False,  $P$ 
23:      end if
24:    end if
25:  end for
26: end for
27: return result
```

Algorithm 4 VerifyNecessarilyPopularMatching

```
1: Input:  $O, N, P = (P_1, \dots, P_n), \mu = \{(1, \sigma(1)), \dots, (n, \sigma(n))\}$ 
2: Output: boolean indicating either the matching is necessarily popular or not.
3:  $cpt_{fpost} := 0$ 
4: for each  $i \in N$  do
5:    $Isfpost := True$ 
6:   // A matching is popular if each agent obtained either its fpost or spost.
7:   // A fpost and/or a spost can not be specified by an agent at the beginning. So we
   check if all the possible completion respect this condition.
8:   while There exist a completion of the agent's  $i$  preference list do
9:      $O_i :=$  the set of non ranked item by agent  $i$ .
10:    for Each unknown  $p \in P_i$  do
11:      Choose an  $o$  in  $O_i$  for each unknown  $p$ 
12:       $O_i \leftarrow O_i \setminus \{o\}$ 
13:    end for
14:    // At the end of this for loop, we get a completion of the agent's  $i$  preference list
15:    // Checking this completion respect the necessarily popular conditions
16:    if not  $(\sigma(i) \in \{f(i), s(i)\})$  then
17:      return False;
18:    else if  $\exists x \in P_i$  such that  $x \succ_i \sigma(i)$  and  $x$  not a fpost then
19:      return False;
20:    end if
21:    if  $\sigma(i)$  is not a fpost in this completion then
22:       $Isfpost \leftarrow False$ 
23:    end if
24:  end while
25:  if  $Isfpost$  then
26:     $cpt_{fpost} \leftarrow cpt_{fpost} + 1$ 
27:  end if
28:  // If at least half of fpost are assigned, the matching  $\mu$  is necessarily popular
29:  if  $cpt_{fpost} \geq \lceil n/2 \rceil$  then
30:    return False;
31:  end if
32: end for
33: return True;
```

4 Examples

Instance for possibly popular matching Let H an instance of the House Allocation problem such as:

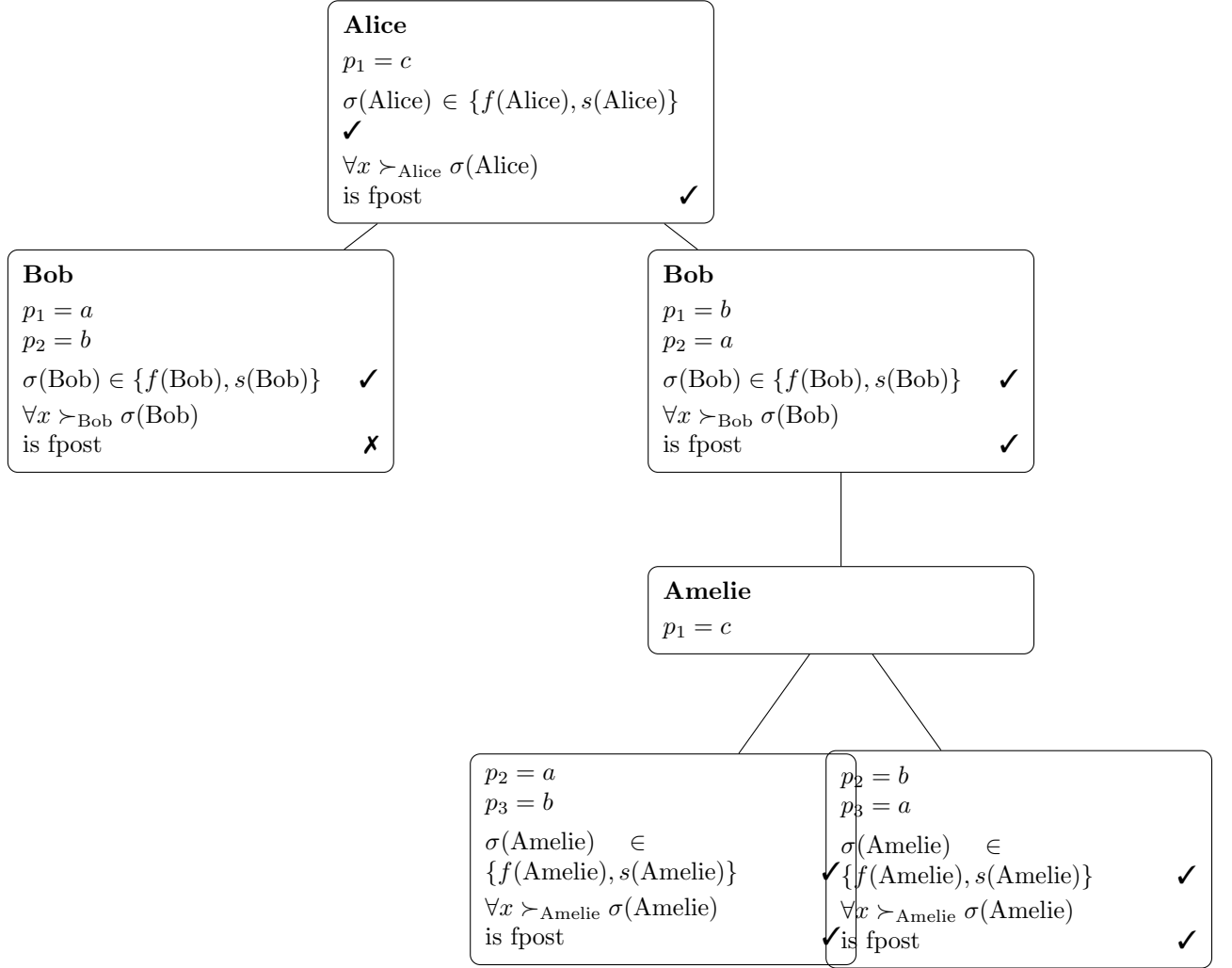
- $N = \{Alice, Bob, Amelie\}$ the set of agents
- $O = \{a, b, c\}$ the set of objects (houses)
- $P = \{P_1, P_2, P_3\}$ the partial profile for which P_i is the strict partial preference order of the agent i .
- $\mu = \{(Alice, b), (Bob, c), (Amelie, a)\}$ the matching. Some object specified in the matching are not ranked in some preference lists.

Agents	P_i	Completions		
Alice	? a ?	$\boxed{b} \textcircled{a} c$	$\boxed{b} a c$	$c a \boxed{b}$
Bob	b ? ?	$\textcircled{b} \boxed{c} a$	$b \boxed{c} a$	$b \boxed{c} a$
Amelie	? a ?	$\textcircled{c} \boxed{a} b$	$b \boxed{a} a$	$b \boxed{a} c$
		not popular	no solution	not popular

Instance for necessarily popular matching - Non necessarily popular matching Let H an instance of the House Allocation problem such as:

- $N = \{Alice, Bob, Amelie\}$ the set of agents
- $O = \{a, b, c\}$ the set of objects (houses)
- $P = \{P_1, P_2, P_3\}$ the partial profile for which P_i is the strict partial preference order of the agent i .
- $\mu = \{(Alice, a), (Bob, b), (Amelie, c)\}$ the matching. Some object specified in the matching are not ranked in some preference lists.

Agents	P_i	Completions		
Alice	? a b	$\boxed{b} \textcircled{a} c$	$\boxed{b} a c$	$c a \boxed{b}$
Bob	? ? a	$\textcircled{b} \boxed{c} a$	$b \boxed{c} a$	$b \boxed{c} a$
Amelie	c ? ?	$\textcircled{c} \boxed{a} b$	$b \boxed{a} a$	$b \boxed{a} c$
		not popular	no solution	not popular



5 Proof

5.1 Proof Verif Possible Popular Matching

5.2 Verification of Possibly Popular Matching

Proposition 6. *Let N be a set of n agents and O a set of n objects. Preferences are given as partial orders (pairwise comparisons), and we consider their completions into strict total orders. Assume a perfect matching setting (each agent is matched to exactly one object and vice versa).*

If there exist $i, j \in N$ such that

$$\sigma(j) \succ_i \sigma(i)$$

and there exists $o \in O$ such that

$$o \succ_j \sigma(j),$$

then μ is not possibly popular.

Definitions

- A **completion** is a strict total order extending the partial preferences.

- A matching μ is **popular** if for every matching μ' :

$$|\{i \in N : \mu(i) \succ_i \mu'(i)\}| \geq |\{i \in N : \mu'(i) \succ_i \mu(i)\}|.$$

- μ is **possibly popular** if it is popular under at least one completion.
- We assume a **perfect matching**: $|N| = |O|$ and all agents/objects are matched.

Proof We prove the contrapositive: assume that there exist $i, j \in N$ and $o \in O$ such that

$$\sigma(j) \succ_i \sigma(i) \quad \text{and} \quad o \succ_j \sigma(j).$$

We show that μ cannot be popular under any completion.

Let μ be any completion of the preferences. We construct a competing matching μ' .

Case 1: o is unmatched in μ .

Define μ' by:

$$\mu'(i) = \sigma(j), \quad \mu'(j) = o,$$

and keep all other assignments unchanged except for maintaining a valid matching.

Case 2: $o = \sigma(k)$ for some agent k .

We define a local exchange (cycle of length 3):

$$i \rightarrow \sigma(j), \quad j \rightarrow o, \quad k \rightarrow \sigma(i).$$

Formally:

$$\mu'(i) = \sigma(j), \quad \mu'(j) = o, \quad \mu'(k) = \sigma(i),$$

and for all other agents $\ell \notin \{i, j, k\}$:

$$\mu'(\ell) = \mu(\ell).$$

Validity: This defines a valid matching since it is a permutation of objects among $\{i, j, k\}$ and leaves all others unchanged.

Locality: The modification from μ to μ' only affects agents i, j, k .

Comparison of preferences:

- Agent i prefers μ' :

$$\mu'(i) = \sigma(j) \succ_i \sigma(i) = \mu(i).$$

- Agent j prefers μ' :

$$\mu'(j) = o \succ_j \sigma(j) = \mu(j).$$

- Agent k may prefer μ or μ' (no guarantee).

- All other agents are indifferent:

$$\mu'(\ell) = \mu(\ell).$$

Vote count:

$$|\{i : \mu'(i) \succ_i \mu(i)\}| \geq 2$$

$$|\{i : \mu(i) \succ_i \mu'(i)\}| \leq 1$$

Thus:

$$|\{i : \mu'(i) \succ_i \mu(i)\}| > |\{i : \mu(i) \succ_i \mu'(i)\}|$$

So μ' is more popular than μ .

Conclusion:

For any completion, we can construct a matching μ' that defeats μ . Hence μ is not popular under any completion.

Therefore, μ is not possibly popular. $\square$

$=_i$

Suppose for a contradiction that the matching μ is popular for a given completion and $\exists i, j \in N$ such that $\sigma(j)P_i\sigma(i)$ and $\exists o \in O$ such that $o \succ_j \sigma(j)$. Given the lemma 2.4 describe in the paper "Popular Matching" of Abraham et al, a matching is popular if and only if each agent is matched either with its fpost or its spost and every fpost is matched.

- The agent i is not matched to its fpost because $\sigma(j) \succ_i \sigma(i)$. So agent i will be matched to its spost and agent j will be matched to the fpost of i .
- The agent j is not matched to its fpost because $\exists o \in O$ such that $o \succ_j \sigma(j)$. So agent j will be matched to its spost and o is its fpost.
- Let μ' be another matching such that $\sigma'(i) = fpost(i) = \sigma(j)$ and $\sigma'(j) = fpost(j) = o$. So agents i and j will prefer μ' to μ and by consequence, μ' is more popular than μ . It is a contradiction because we have assumed that μ is popular.

To conclude, this proves the lemma.

$\models$ Suppose that the matching μ is not popular. So $\exists i \in N$ such that $\sigma(i) \notin \{fpost(i), spost(i)\}$. By consequence, $\exists j, k \in N$ such that $\sigma(j) = fpost(i)$ and $\sigma(k) = spost(i)$.

1. Either $\sigma(j) = fpost(j)$ and $\sigma(k) = fpost(k)$. It is a contradiction because μ is not popular ($|\{i \in N : \sigma(i) \succ_i \sigma'(i)\}| < |\{i \in N : \sigma'(i) \succ_i \sigma(i)\}|$).
2. Either one of them, e.g k gets its fpost and j is matched with its spost. So $\exists o \in O$ such that $o \succ_j \sigma(j)$ and $\sigma(j) \succ_i \sigma(i)$. The property is verified.
3. Either one of them, e.g k gets its fpost and $\sigma(j) \notin \{fpost(j), spost(j)\}$. So $\exists o \in O$ such that $o \succ_j \sigma(j)$ and $\sigma(j) \succ_i \sigma(i)$. The property is verified.

This proves the reciprocity lemma.

Let's try to prove this lemma by contradiction. Supposing the property **Prop**: $\exists i, j \in N$ such that $\sigma(j)P_i\sigma(i)$ and $\exists o \in O$ such that $o \succ_j \sigma(j)$ is true. From this property, we assume the matching μ not possibly popular.

A matching μ is possibly popular if it is popular in at least one completion of the partial preference profile.

Initialization For 2 agents and 2 objects:

$$i : \sigma(j)P_i\sigma(i)$$

$$j : \sigma(i)P_j\sigma(j)$$

$\exists \mu' = \{(i, \sigma(j)), (j, \sigma(i))\}$ such that $|\{i \in N : \sigma(i) \succ_i \sigma'(i)\}| < |\{i \in N : \sigma'(i) \succ_i \sigma(i)\}|$ (0 < 2). Hence μ is not a popular matching whatever the completion (there are only two possible completions for this setting) for 2 agents and 2 objects.

Inheritance We suppose the property Prop true for n agents and n objects and we want to determine whether the property is still true for $n+1$ agents and $n+1$ objects. If μ is not popular at rank n , this means there is no completion such that $|\{i \in N : \sigma(i) \succ_i \sigma'(i)\}| \geq |\{i \in N : \sigma'(i) \succ_i \sigma(i)\}|$. Let's add an agent and an object: By contraposition, we have

$$A \Rightarrow B$$

$$not B \Rightarrow not A$$

where

B: $\forall$ completions over $n + 1$, μ is not popular.

A: $\forall$ completions over n , μ is not popular.

not B: $\exists$ a completion over $n + 1$, such that μ is popular.

not A: $\exists$ a completion over n , such that μ is popular.

In other words, if we suppose there exists a completion over $n + 1$ such that μ is popular, then this implies that there exists a completion over n such that μ is popular. However, we said that μ is not popular over n so all completions over $n + 1$ make μ not popular.

Nevertheless, we must show that if $\exists$ a completion over $n + 1$ such that μ is popular then its restriction over n is still popular.

Over $n + 1$: $|\{i \in N : \sigma(i) \succ_i \sigma'(i)\}| \geq |\{i \in N : \sigma'(i) \succ_i \sigma(i)\}| \forall \mu'$ We restraint by removing 1 agent and 1 object (called agent k and object k):

1. agent $k \in \{i \in N : \sigma(i) \succ_i \sigma'(i), \forall \mu'\}$ and object k is not $\sigma(k)$
2. agent $k \in \{i \in N : \sigma(i) \succ_i \sigma'(i), \forall \mu'\}$ and object k is $\sigma(k)$
3. agent $k \in \{i \in N : \sigma'(i) \succ_i \sigma(i), \forall \mu'\}$ and object k is not $\sigma(k)$
4. agent $k \in \{i \in N : \sigma'(i) \succ_i \sigma(i), \forall \mu'\}$ and object k is $\sigma(k)$